

Summary Tables and Workflow Diagram

Table 1: Tasks and Key Concepts

Task	Main Tool / Method	Key Concept Learned
Task 1	Github, venv, Linux WSL	Repository hygiene, isolated environments, and navigating the terminal.
Task 2	Python fundamentals	Functions, type hints, file I/O, and exception handling.
Task 3	Manual parser + pandas	How CSV text becomes structured data; why libraries handle edge cases that manual parsing cannot.
Task 4	pandas data cleaning	Identifying and fixing duplicates, type errors, missing values, and inconsistent categories in real data.
Task 5	matplotlib	Producing labeled, saved charts from cleaned data.

Table 1 — Summary of Tasks 1–5 and the main concept learned from each task.

Figure 1: Data Pipeline — From Raw CSV to Visualized Insights

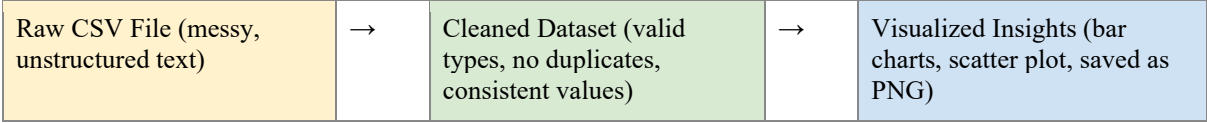


Figure 1 — Workflow from raw CSV through cleaning to final plots.